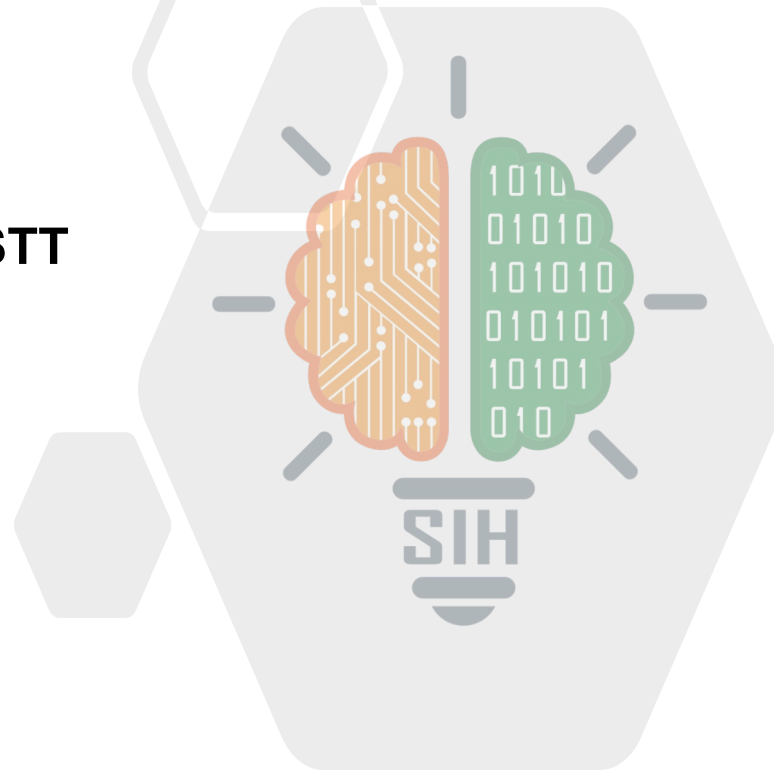


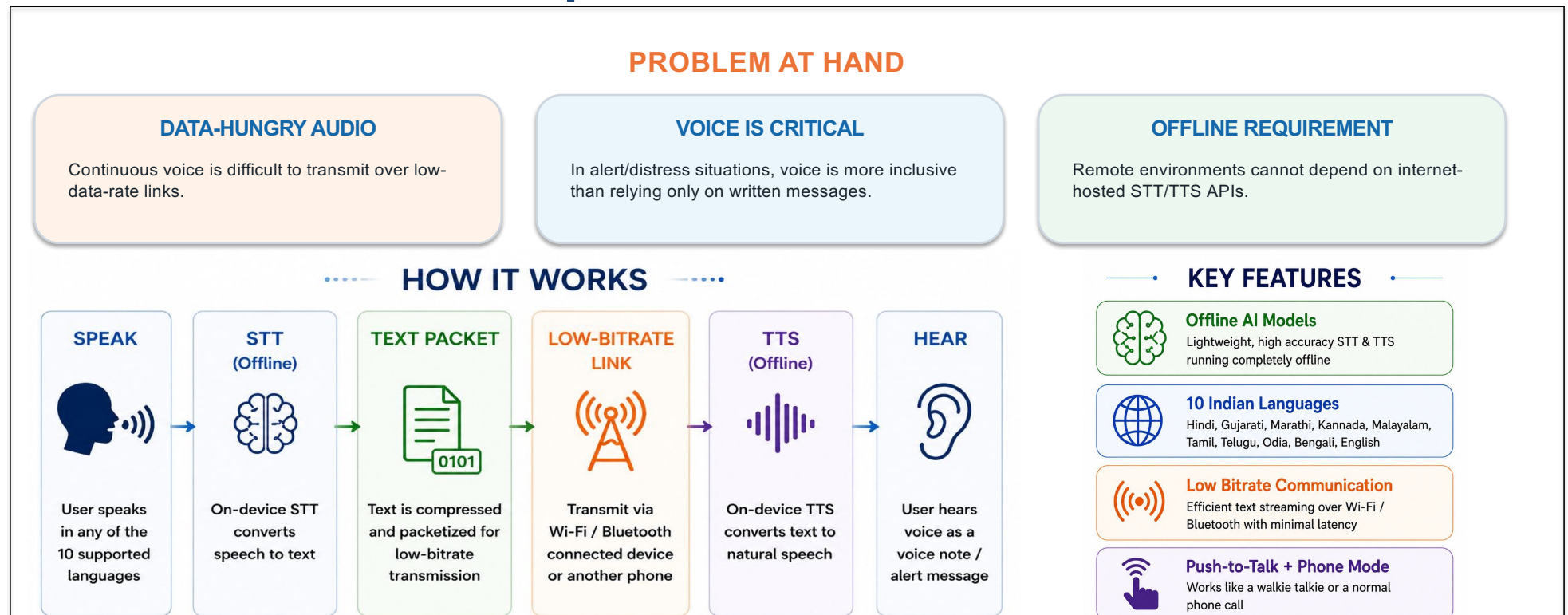
SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26173**
- **Problem Statement Title-**
iTantra -Indian Multilingual TTS& STT
Aided Neural Transceiver Radio
Access for low bitrate links
- **Theme- Smart Automation**
- **PS Category - Software**
- **Team ID-**
- **Team Name (Registered on portal) - Odyysix**



Proposed Solution



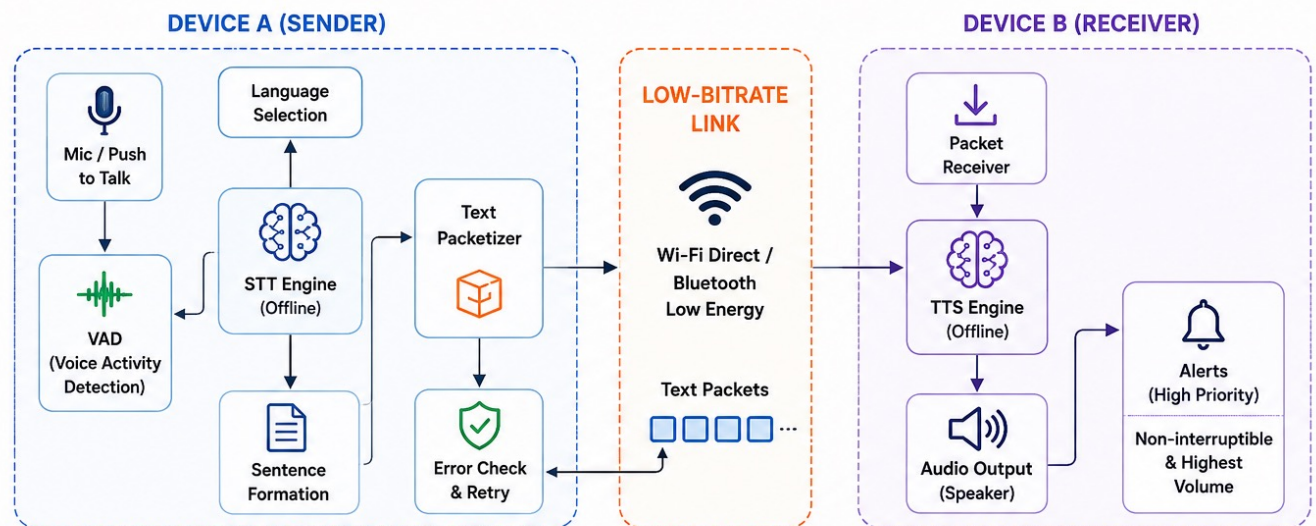
ON-DEVICE AI

- Open-source STT/TTS pipeline
- Local inference — no hosted speech API
- Model optimization for low-/mid-range Android
- Target: low STT Word Error Rate + intelligible TTS

COMMUNICATION

- Convert speech to compact text
- Packetize and transmit sentence data
- Bluetooth / Wi-Fi Direct for local demo
- Modular interface for low-bitrate links

SYSTEM ARCHITECTURE



FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

- Android application can host optimized open-source speech models.
- Offline inference directly satisfies the PS requirement.
- Bluetooth / Wi-Fi Direct provides a practical two-phone demonstration path.
- Prototype can be built incrementally: 1–2 languages → full language set.

RESOURCE FEASIBILITY

- Use lightweight models and optimize memory/CPU footprint.
- Sentence-level text transmission greatly reduces payload compared with raw audio.
- Use asynchronous STT → transmission → TTS to reduce perceived delay.

KEY CHALLENGES

- STT accuracy across Indian languages, accents and noisy environments.
- Model size and CPU/RAM limits on low- and mid-range phones.
- Low end-to-end latency while maintaining intelligibility.
- Reliable packet handling when the communication link is constrained.

MITIGATION PLAN

- Quantization / lightweight inference and model profiling.
- Voice Activity Detection + pause-based sentence formation.
- Streaming/incremental processing where supported.
- Buffering, packet framing and recovery for the communication layer.

IMPACT AND BENEFITS



Bridges Language Gaps

Enables real-time voice communication across 10 Indian languages without internet.



Works Offline

Fully on-device STT & TTS ensures communication even in no-network environments.



Low Data Usage

Uses low-bitrate text streaming instead of audio, saving bandwidth and reducing cost.



Easy & Inclusive

Push-to-talk + phone-mode makes it simple for anyone to use, regardless of age or literacy.



Reliable & Safe

Critical alerts are loud, non-interruptible and reach users instantly in any situation.



Empowers Everyone

Supports emergency teams, field workers, travelers and communities for seamless communication.



Saves Lives

Faster communication during emergencies and critical situations.



Cost Effective

Minimal data usage and works on low-cost devices.



Reaches Everywhere

Effective in remote, rural, and disaster-prone areas.



Stronger Communities

Builds a more connected, inclusive and safer society.



AI4Bharat IndicConformer ASR – GitHub

Indian language ASR models (22 official languages)
<https://github.com/AI4Bharat/IndicConformerASR>



AI4Bharat Indic-TTS – GitHub

Indian language TTS models (13 languages)
<https://github.com/AI4Bharat/Indic-TTS>



Mozilla Common Voice – Dataset

Open speech dataset for ASR/TTS evaluation
<https://commonvoice.mozilla.org/data>



Silero VAD – GitHub

Lightweight Voice Activity Detection for speech segmentation
<https://github.com/snakers4/silero-vad>



Android LiteRT / Custom ML – Documentation

On-device ML deployment for Android
<https://developer.android.com/ai/custom>



Android Wi-Fi Direct – Documentation

Peer-to-peer device communication (Offline)
<https://developer.android.com/develop/connectivity/wifi/wifi-direct>

REFERENCES

- 1) L. R. Rabiner and R. W. Schafer, *Digital Processing of Speech Signals*, Prentice-Hall, 1978.
- 2) L. R. Rabiner and B. H. Juang, *Fundamentals of Speech Recognition*, Prentice Hall, 1993.
- 3) AI4Bharat, *IndicConformer: Conformer-based ASR for Indian Languages*, 2024.
- 4) A. Sankar et al., “IndicVoices-R: Unlocking a Massive Multilingual Multi-speaker Speech Corpus for Scaling Indian TTS,” *NeurIPS 2024 Datasets and Benchmarks*, 2024.